EXERCISE 1 [4 marks]

Find the equation of the tangent to the curve

$$y = e^{3x^2+5x-2}$$

at the point $(-2, 1)$, giving your answer in the form $ax + by + c = 0$, where a, b, c are integers.

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EXERCISE 2 Consider the function f defined by

$$f(x) = xe^{3\cos x}, \quad \text{for } -\pi \leq x \leq \pi.$$

1. Find the number of points at which the graph of f has a horizontal tangent. [1 mark]

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The point A is the point on the graph of f for which the x-coordinate is $\frac{\pi}{2}$.

2. Show algebraically that the gradient of the tangent to the graph of f at point A is

$$\frac{2-3\pi}{2}.$$

[4 marks]

3. Hence find the equation of the normal line to the graph of f at point A, and determine where that line intersects the x-axis. [5 marks]

4. Show algebraically that the graph of f intersects the line $y = x$ in exactly three places, and determine the coordinates of the points of intersection. [4 marks]